Human – Computer Interaction

Principle of Human Computer Interaction Faculty of Computer Science and Information Technology

Lecture: 1

Introduction to HCI

- Interacting with technology has become an essential part of everyday life for the majority of people.
- The average user of a computer system is now less likely to understand the technology. Since, there are different types of technology they have to use.
- People are busy and may spend little or no time actually learning a new system.

Introduction…(cont.)

- Therefore, computer systems should be easy to use, easy to learn, and with no errors.
- To design and develop of such a system is a major concern of HCI

What is HCI?

- Human-computer interaction (HCI): “is a discipline concerned with the design, evaluation and implementation of interactive systems for human use and with study of major phenomena surrounding them.” (ACM SIGCHI, 1992,

What ..(cont.)

HCI consists of three parts:

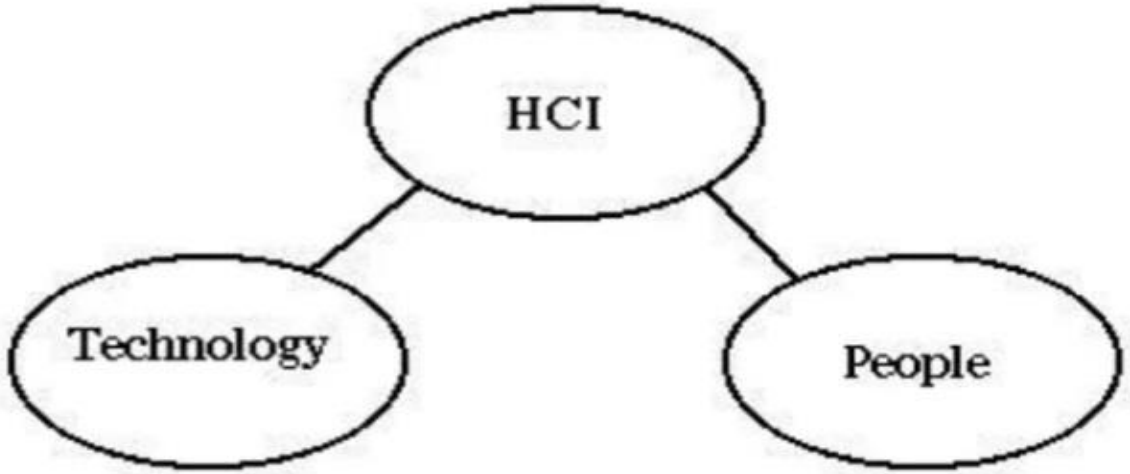
- Human: could be an individual user or a group of users.
- Computer: could be any technology ranging from the general desktop computer to a large-scale computer system.
- Interaction: any direct or indirect communication between a human and computer.

What ..(cont.)

- HCI concentrates on the study of human factors
- The study of human factors started during the Second World War by US army.
- Usability was born because of badly designed arms that caused "friendly fire" during war.

The HCI Challenge

The HCI Challenge



What can it do?
How can it be built?
What are the possibilities?

What are people doing?
How would it fit it?
What would they do with the technology?

HCI is Not about

- Making the interface look pretty
- Only about desktop computers (and that goes for computing as well!)
- Something that would be nice to do but usually there's no time for it

HCI is Not about

- Understanding the users
- Understanding users tasks
- Understanding the surrounding environment
- GUI requirements gathering and analysis
- Design prototype
- Evaluate the system

The goals of HCI

- The goal of HCI “is to develop or improve the safety, utility, effectiveness, efficiency and usability of system that include computers.” (Interacting with computers, 1989, p3)

The goals of HCI

• The goals of HCI are to produce usable and safe systems, as well as functional systems. In order to fulfill that, developers must attempt to:

- Understand how people use technology
- Building suitable systems
- Achieve efficient, effective, and safe interaction
- Put people first.

People needs, capabilities and preferences should come first. People should not have to change the way that they use a system. Instead, the system should be designed to match their requirements

What is Usability?

- A usable system is:
  - easy to use
  - easy to learn
  - easy to remember how to use
  - effective to use
  - efficient to use
  - safe to use
  - enjoyable to use

Why is usability important?

A Good user-interface can:

1. Earn a company billions.
2. Increase users loyalty.
3. Increase users trust.
4. Makes users happy :)

Why ..(cont.)

A Bad user-interface can:

1. be annoying, embarrassing, frustrating, and even deadly.
2. Increase mistakes in data entry and system operation.
3. Makes functions become completely inaccessible.
4. System failure because of user rejection.

Why HCI is Important in the Context of WWW?

- Competition is very close (just another link…)
- Comparison is easily possible (example – Online-Shop)
- Users who can’t find the product in the shop can not buy it
- Users who are not able to fill in correctly the order form are not going to buy

It is not Simple to Make GoodUser Interfaces

Basic misconceptions:

- If I (the developer) can use it, everyone can use it
- If our non-technical staff can use it, everyone can
- Good user interfaces are applied common sense
- A system is usable if all style guidelines are met

Examples of good and bad design

Examples of good and bad design

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